Richard Alexander

School of Physics & Astronomy, University of Leicester

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Employment

2009 –	**School of Physics & Astronomy, University of Leicester** Professor of Theoretical Astrophysics (2019–) Reader (Associate Professor) in Theoretical Astrophysics (2015–2019) Lecturer in Theoretical Astrophysics (2009–2015) STFC Advanced Fellow (2009–2015) [Paternity leave: March–June 2012]
2007 – 2009	**Sterrewacht Leiden, Leiden University** Postdoctoral Fellow
2005 – 2007	**JILA, University of Colorado** Postdoctoral Research Associate

Education

2002 – 2005	**Institute of Astronomy, University of Cambridge** PhD in Astrophysics Supervisor: Prof. Cathie Clarke
1997 – 2002	**University of Edinburgh** MPhys (Astrophysics) with First Class Honours

Research Supervision

PhD students	Simin Tong (2022–26); Giulia Ballabio (2017–20); Tom Hands (2012–16); Alex Dunhill (2010–13)
Postdocs	Sahl Rowther (2022–26); Alison Young (2019–21); Andrew Winter (2019–20); Enrico Ragusa (2018–21); Rebecca Nealon (2017–20); Cassandra Hall (2017–20); Giovanni Dipierro (2017–19); Stephen Fendyke (2014–16)
Pre-doctoral	Sarah Crick (née Smedley), SURE summer research student (2010); Daniel Harsono, MSc research project (Leiden University; 2008–09)

Current Teaching & Lecturing

2022 –	Lecturer & module lead (2023–) for 4th-year astrophysics course *SMBHs & Exoplanets*
2021 –	Departmental postgraduate tutor (i.e., PhD programme director)
	Lecturer & module lead (2022–) for 3rd-year astrophysics course *Quasars & Cosmology*
2015 –	Lecturer & module lead (2017–) for 1st-year core physics course *Waves & Quanta*
2011 –	Module lead for postgraduate course *Formation of Planetary Systems*, University of Leicester & Midlands Physics Alliance Graduate School (MPAGS)

Research Grants (as Principal Investigator)

2022 – 2026	**Science & Technology Facilities Council (UK)** PI of funded project (£525k) within departmental Consolidated Grant (£1.5M total)

Research Grants (as Principal Investigator) (continued)

2016 – 2021	**European Research Council (EU)** Consolidator Grant (€1.95M, PI)
2014 – 2017	**The Leverhulme Trust (UK)** Philip Leverhulme Prize (£70k, PI)
2013 – 2016	**Science & Technology Facilities Council (UK)** PI of funded project (£154k) within departmental Consolidated Grant (£4.8M total)
2009 – 2015	**Science & Technology Facilities Council (UK)** Advanced Fellowship (£498k, PI)

Large Facility Awards

- Competitively-allocated large awards on the NCSA (USA), NCF (Netherlands) and DiRAC (UK) high-performance computing facilities, totalling over 60 million CPU hours.
  Recent highlights include PI awards on DiRAC in 2021, 2023 & 2025.
- More than 20 competitively-allocated observing time awards on international telescopes.
  Recent highlights include large programmes on the ESO Very Large Telescope and the Hubble Space Telescope (Odysseus), and projects on JWST and ALMA.

Service & Peer Review

- Grant panels: Royal Society University Research Fellowships (2025–); RAS Fellowships (2026–); STFC Ernest Rutherford Fellowships (2020–2024); PRACE–DECI (2020); NASA *Origins of Solar Systems* (2010, 2013)
- Referee for *Science*, *Nature*, *ApJ*, *ApJL*, *ApJS*, *MNRAS*, *AJ*, *A&A*, *New Astronomy Reviews*, *Space Science Reviews*: typically 4–6 articles per year.
- Review of grant applications for STFC, UKRI, Royal Society, RAS, Leverhulme Trust (UK), ERC (EU), NASA (USA), NWO (Netherlands), FWF (Austria), CSCS (Switzerland), CONICYT (Chile).

Invited Talks (selected)

2026	*UK & Ireland Discs Meeting 2026*, Cambridge, UK (keynote review)
2024	*Formation of the protosolar disk & its first planetesimals*, Collège de France (invited lecture)
2023	*Core-to-Disk III*, Saclay, France (keynote review)
2020	*The Solar-Stellar Connection*, Warwick, UK (summer school lecturer)
2018	*Where are the young planets?*, Madrid, Spain (keynote review)
2017	*The migration issue*, Cambridge, UK (conference-opening review)
2016	*Star formation 2016*, Exeter, UK (keynote review)
2014	*Protoplanetary disks and planets*, Santiago, Chile (keynote review)
2013	*Protostars & Planets VI*, Heidelberg, Germany (PI of review chapter)

Publication & Citation Summary

- 93 peer-reviewed papers (published or in press).
- 21 peer-reviewed papers as lead author.
- 29 peer-reviewed papers led by students and postdocs under my supervision.
- $>$5000 total citations; h-index of 42.